

ADAPT-VQE fails with Aer simulator

<https://github.com/qiskit-community/qiskit-aqua/issues/1467>

ROW 87

ADAPT-VQE fails with Aer simulator #1467



Closed



MariaSapova opened on Dec 2, 2020

...

Information

- Qiskit Aqua version: 0.23.1
- Python version: 3.6.8
- Operating system: Ubuntu

What is the current behavior?

Adapt-VQE works fine with BasicAer, but fails with Aer backends (both statevector_simulator and qasm_simulator). In previous qiskit version VQEAdept worked fine.

```
AquaError: 'Circuit execution failed: ERROR: Failed to load qobj: Invalid parameterized qobj: instructio
```



Steps to reproduce the problem

After molecule definition I use built-in methods:

```
backend = Aer.get_backend('qasm_simulator')
quantum_instance = QuantumInstance(backend)
solver = VQEUCCSDFactory(quantum_instance)
adapt = AdaptVQE(transformation, solver)
res = adapt.solve(driver)
```



What is the expected behavior?

Computation runs fine without error, as in BasicAer case

Assignees

No one assigned

Labels

type: bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Ensure aux_operator eigenvalues are normalized
qiskit-community/qiskit-aqua

Notifications

Customize

Subscribe

You're not receiving notifications from this thread.

Participants



mrossinek on Dec 3, 2020 · Hidden as outdated

Edits



...

mrossinek added status: can't reproduce status: needs informa... on Dec 3, 2020



mrossinek on Dec 3, 2020

Member

...

Having found out about the meta package version numbering I know was able to reproduce your error with a stable-branch installation of Qiskit. Let me dig into this a little bit more and get back to you on this.

mrossinek added status: confirmed type: bug and removed status: can't reproduce status: needs informa... on Dec 3, 2020



mrossinek on Dec 3, 2020

Member

...

PR #1447 fixes this issue and was merged about a week ago. Thus, my initial attempt of reproducing the bug failed, because it is no longer present on the master branch.

This means that you have several options:

- wait for the next meta-package release
- install Qiskit Aqua manually at the commit 8583056 (even 0bf9873 is fine, but the commit after that will not work unless you install Terra from the master branch, too)

We can also see whether it is possible to release a Qiskit Aqua 0.8.2 based on [0bf9873](#) which would allow you to install this bug fix via pip but that is not a decision which I can make. [@woodsp-ibm](#) [@manoelmarques](#) what do you think about this?

 [mrossinek](#) removed status: confirmed on Dec 3, 2020



[MariaSapova](#) on Dec 4, 2020

Author ...

[@mrossinek](#) thank you! I installed master version of qiskit and now it works fine. However, I would like to point out one thing: with qasm simulator `aux_operator` values grow quadratically with the number of shots. I've already mentioned the same thing for QEOM #1460.

I think that it would be nice to normalize them.

```
In [6]: backend = Aer.get_backend('qasm_simulator')
quantum_instance = QuantumInstance(backend, shots = 1000, noise_model = None)

optimizer = SLSQP(maxiter = 200, disp = True)

solver = VQEUCSDFactory(quantum_instance, include_custom = True)
adapt = AdaptVQE(qubitOp_transformation, solver)
res = adapt.solve(driver)
print(res)

=== GROUND STATE ENERGY ===

* Electronic ground state energy (Hartree): -1.857275030145
  - computed part:      -1.857275030145
  - frozen energy part: 0.0
  - particle hole part: 0.0
~ Nuclear repulsion energy (Hartree): 0.719968994449
> Total ground state energy (Hartree): -1.137306035696

=== MEASURED OBSERVABLES ===

0: # Particles: 1948676.000 S: 0.000 S^2: 0.000 M: 0.000

=== DIPOLE MOMENTS ===

~ Nuclear dipole moment (a.u.): [0.0 0.0 1.3889487]

0:
* Electronic dipole moment (a.u.): [0.0 0.0 1353305.49997601]
  - computed part:      [0.0 0.0 1353305.49997601]
  - frozen energy part: [0.0 0.0 0.0]
  - particle hole part: [0.0 0.0 0.0]
> Dipole moment (a.u.): [0.0 0.0 -1353304.11102731] Total: 1353304.11102731
                      (debye): [0.0 0.0 -3439755.62367468] Total: 3439755.62367468
```



[woodsp-ibm](#) on Dec 4, 2020

Member ...

While you seem to have an appropriate ground state the other operators, such as number of particles (1.948676 million!) and dipole moment are way off (to put it mildly!) [@mrossinek](#) Any comments here?

 [mrossinek](#) mentioned this on Dec 4, 2020

🔴🔴 Ensure `aux_operator` eigenvalues are normalized #1473



[mrossinek](#) on Dec 4, 2020

Member ...

Hi [@woodsp-ibm](#). I was actually just working on a fix together with [@Cryoris](#). See the new PR [#1473](#)

While you seem to have an appropriate ground state the other operators, such as number of particles (1.948676 million!) and dipole moment are way off (to put it mildly!) @mrossinek Any comments here?

 mrossinek mentioned this on Dec 4, 2020

❗ Ensure aux_operator eigenvalues are normalized #1473



mrossinek on Dec 4, 2020

Member ...

Hi @woodsp-ibm. I was actually just working on a fix together with @Cryoris.
See the new PR #1473

 mrossinek added 2 commits that reference this issue on Dec 4, 2020

Ensure aux_operator eigenvalues are normalized ...

Verified ffc7b1a

Ensure aux_operator eigenvalues are normalized ...

Partially verified e72216e

 Cryoris linked a pull request that will close this issue ❗ Ensure aux_operator eigenvalues are normalized #1473 on Dec 5, 2020

 mrossinek added a commit that references this issue on Dec 9, 2020

Ensure aux_operator eigenvalues are normalized ...

Partially verified 49099e2

 mrossinek mentioned this on Dec 9, 2020

🔗 Ensure aux_operator eigenvalues are normalized Qiskit/qiskit#5496



mrossinek on Dec 10, 2020

Member ...

Since the Aqua core is in the process of being merged into Terra, I re-opened the PR which fixes this over there. See Qiskit/qiskit#5496

 mergify added a commit that references this issue on Dec 28, 2020

Ensure aux_operator eigenvalues are normalized (#5496) ...

Partially verified 4d3ed6a

 molar-volume added a commit that references this issue on Dec 29, 2020

Ensure aux_operator eigenvalues are normalized (Qiskit#5496) ...

4e8a076



mrossinek on Jan 4, 2021

Member ...

This issue is now also fixed in the migrated Aqua core inside of Terra. Closing.

 mrossinek closed this as **completed** on Jan 4, 2021

 mergify added a commit that references this issue on Jan 13, 2021

Improved handling of ``params`` argument in ``Gradient.convert`` (#55) ...

Partially verified 799ad14

Ensure aux_operator eigenvalues are normalized #5496

<> Code

Merged

 mergify merged 5 commits into `Qiskit:master` from `mrossinek:ensure-normalized-aux-op-values-in-vqe` on Dec 28, 2020

Conversation 1

Commits 5

Checks 0

Files changed 2

+20 -1

mrossinek commented on Dec 9, 2020

Member

Summary

This migrates [qiskit-community/qiskit-aqua#1473](#) to Terra, now that the Aqua core has been merged.

As reported on multiple occasions in Qiskit Aqua (<https://github.com/Qiskit/qiskit-aqua/issues/1460>, [qiskit-community/qiskit-aqua#1467](#)), the eigenvalues of the auxiliary operators are not correctly normalized when the QASM backend is used.

Details and comments

This commit fixes this short-coming by applying the same normalization to the VQE's eigenstate when obtained from a QASM backend (in which case this is a dictionary) as done by the `CircuitSampler` in its `sample_circuits` function.

The case of the statevector backend is unaffected by this change, as it will return the eigenstate as a list.

A unittest is added which asserts this behavior.

Co-authored-by: Julien Gacon gaconju@gmail.com

1

Reviewers

ikkoham

✓

Cryoris

✓

manoelmarques

woodsp-ibm

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Ensure aux_operator eigenvalues are normalized

Partially verified

49899e2

mrossinek requested a review from **a team** as a code owner 5 years ago

mrossinek mentioned this pull request on Dec 10, 2020

ADAPT-VQE fails with Aer simulator [qiskit-community/qiskit-aqua#1467](#)

Closed

ikkoham approved these changes on Dec 11, 2020

View reviewed changes

ikkoham left a comment

Contributor

Thank you for the previous PR. LGTM.

Notifications

Customize

Subscribe

You're not receiving notifications from this thread.

3 participants